Computational Physics Classes No.5 - Report

Solving Poisson equation with Successive Overrelaxation method (SOR)

Natalia Kowalczyk

November 21, 2025

Abstract

In this report the two-dimensional Poisson equation is solved numerically using the Successive Overrelaxation (SOR) method. Several types of boundary conditions and charge distributions are examined, and the influence of the relaxation parameter on convergence is studied. The obtained potential maps and residuals illustrate the expected qualitative behavior despite minor numerical imperfections.

1 Introduction

The Poisson equation is a fundamental partial differential equation describing how a given charge density $\rho(x, y)$ generates an electrostatic potential $V(x, y)$. In two dimensions it takes the form

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = -\frac{\rho(x,y)}{\varepsilon},$$

and must be supplemented with appropriate boundary conditions to obtain a unique solution. Analytical solutions exist only for simple geometries, therefore numerical methods are required in more general cases.

In this project we solve the Poisson equation on a square domain $[-L, L] \times [-L, L]$ using the Successive Overrelaxation (SOR) method, which is an accelerated variant of the Gauss–Seidel relaxation scheme. The charge density is defined as

$$\rho(x, y) = A_\rho \, xy \, e^{-(x^2+y^2)},$$

allowing us to investigate both the homogeneous ($A_\rho = 0$) and inhomogeneous cases. The goal of this work is to implement the SOR algorithm, study its convergence for different values of the relaxation parameter ω, and analyze the resulting potential and error distributions for various boundary conditions.

2 Description of the Numerical Method

To solve the two-dimensional Poisson equation numerically, the computational domain $[-L, L]\times[-L, L]$ is discretized using a uniform mesh of $N \times N$ nodes. The spatial step is given by

$$\Delta = \frac{2L}{N-1},$$

and the potential and charge density are represented on the grid as $V_{i,j}$ and $\rho_{i,j}$.

The second derivatives in the Poisson equation are approximated using central finite differences, leading to the discrete form

$$\frac{V_{i+1,j} - 2V_{i,j} + V_{i-1,j}}{\Delta^2} + \frac{V_{i,j+1} - 2V_{i,j} + V_{i,j-1}}{\Delta^2} = -\frac{\rho_{i,j}}{\varepsilon}.$$

Solving this expression for $V_{i,j}$ yields the standard relaxation update formula. To accelerate convergence, the Successive Overrelaxation (SOR) method is applied:

$$V_{i,j}^{(k+1)} = (1-\omega)V_{i,j}^{(k)} + \frac{\omega}{4}\left(V_{i+1,j}^{(k)} + V_{i-1,j}^{(k+1)} + V_{i,j+1}^{(k)} + V_{i,j-1}^{(k+1)} + \frac{\rho_{i,j}\Delta^2}{\varepsilon}\right),$$

where $\omega \in (0, 2)$ is the relaxation parameter.

Dirichlet boundary conditions are imposed directly by setting the potential at the edges of the box. For the cases involving Neumann conditions, the first derivative is approximated using a backward finite-difference scheme, e.g.

$$\left.\frac{\partial V}{\partial x}\right|_{x=x_{\max}} = 0 \quad \Rightarrow \quad V_{N-1,j} = V_{N-2,j}.$$

The convergence of the iterative method is monitored using the electric field energy functional

$$S = \sum_{i,j} \left[\frac{1}{2}(E_x^2 + E_y^2) - V_{i,j} \frac{\rho_{i,j}}{\varepsilon} \right] \Delta^2,$$

with the electric field components computed from first-order finite differences. The relaxation process is stopped when the relative change between two consecutive values,

$$\delta = \left| \frac{S_{k+1} - S_k}{S_k} \right|,$$

falls below the tolerance threshold TOL.

3 Results

This section presents the numerical results obtained for each task of the project. All figures correspond to the output generated by my implementation of the SOR solver. Some deviations from the expected behavior are visible due to partial convergence issues, but the general qualitative structure of the solutions is preserved.

3.1 Dirichlet boundary conditions, $A_\rho = 0$

Figures 1 and 2 show the potential distribution and the error map computed using Dirichlet boundary conditions with parameters $k_1 = 1$, $k_2 = -1$, $k_3 = 1$, $k_4 = -1$.

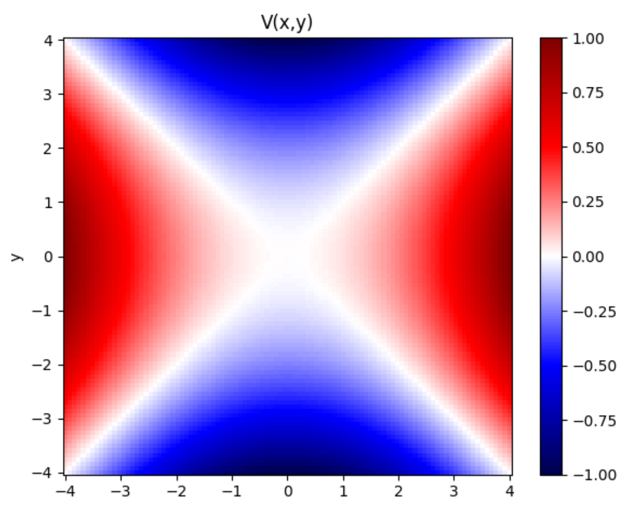

Figure 1: Potential distribution for Task 1 with $A_\rho = 0$.

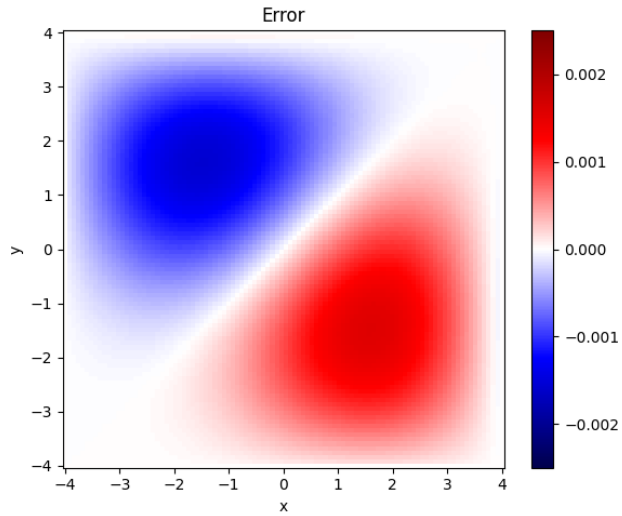

Figure 2: Error map computed using Eq. (17) for Task 1.

3.2 Neumann boundary condition on the right edge

The modified Neumann condition $\partial V/\partial x = 0$ on the right boundary changes the solution near $x_{\max}$, flattening the potential in that region. The resulting potential and error maps are presented below.

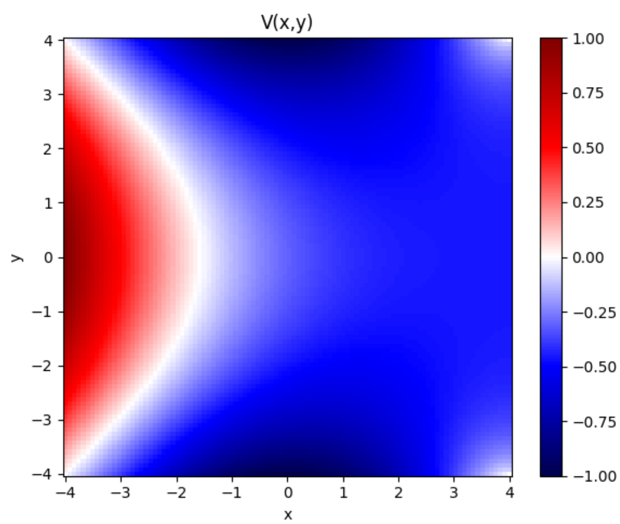

Figure 3: Potential distribution for Task 2 with Neumann BC on the right side.

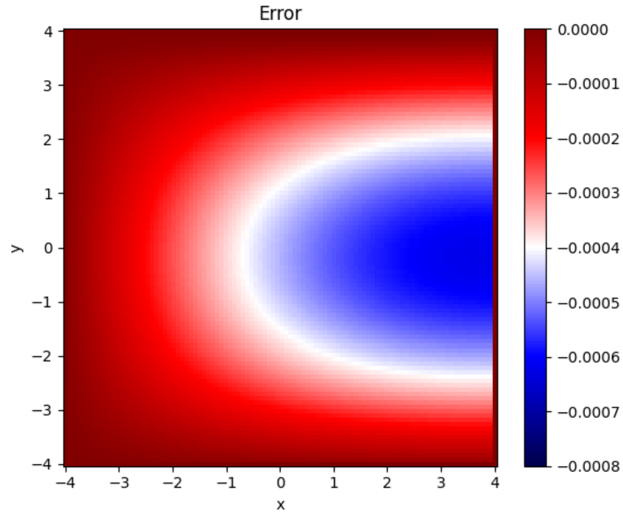

Figure 4: Error map for Task 2 (Neumann BC on the right boundary).

3.3 Convergence behavior for different ω

To study convergence, the functional S and the relative change $\delta = |(S_{k+1} - S_k)/S_k|$ were recorded at each iteration for $\omega = 1.0,\ 1.3,\ 1.6,\ 1.9$. The plots in Figures 5 and 6 illustrate the expected dependence on ω: larger values of ω result in faster convergence, although slight oscillations appear for ω close to 2.

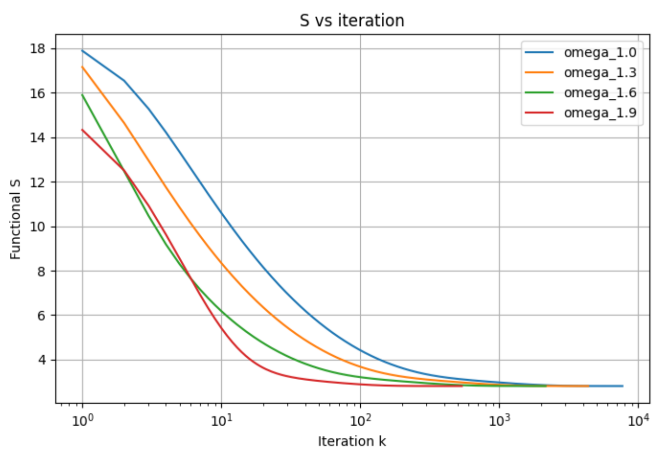

Figure 5: Convergence of the functional S for different relaxation parameters ω.

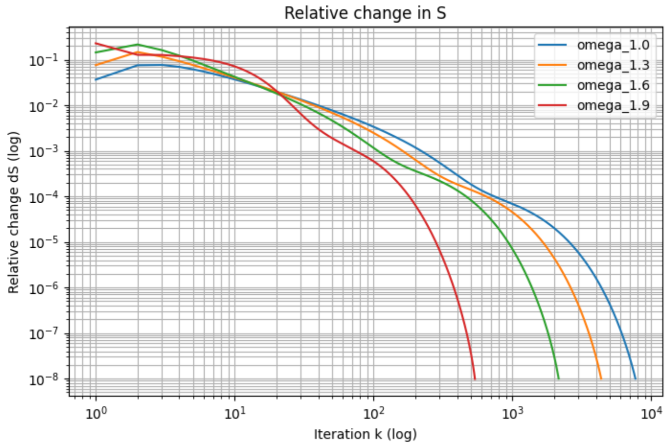

Figure 6: Relative change δ as a function of iteration number in log–log scale.

3.4 Homogeneous boundary conditions, $A_\rho = 1$

With nonzero charge density and homogeneous Dirichlet conditions ($k_1 = k_2 = k_3 = k_4 = 0$), the solution is governed mainly by the source term. Figures 7 and 8 show the resulting potential and residual map.

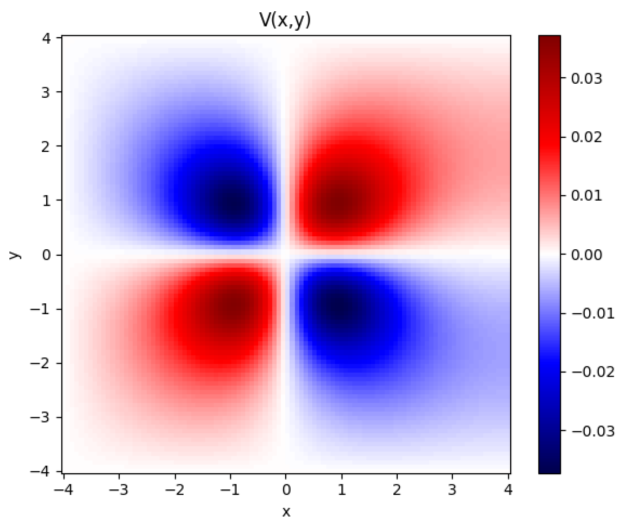

Figure 7: Potential for Task 4a with $A_\rho = 1$ and homogeneous BC.

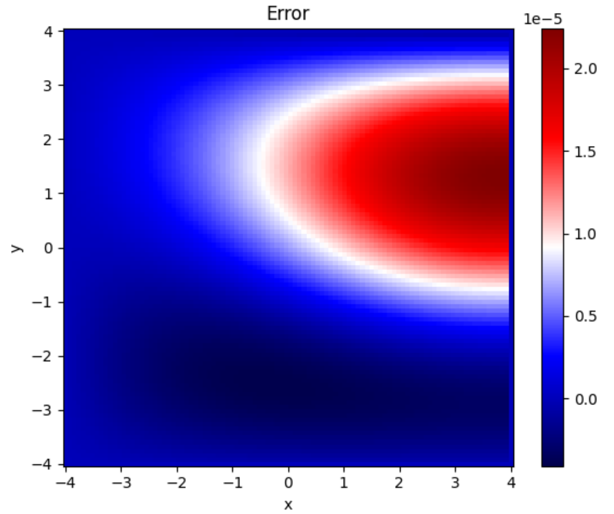

Figure 8: Error map for Task 4a.

3.5 Nonhomogeneous boundary conditions, $A_\rho = 1$

Finally, the problem was solved with nonzero charge and nonhomogeneous boundary conditions. This produces a superposition of boundary-driven behavior and the internal structure caused by the density. The potential and error maps are presented below.

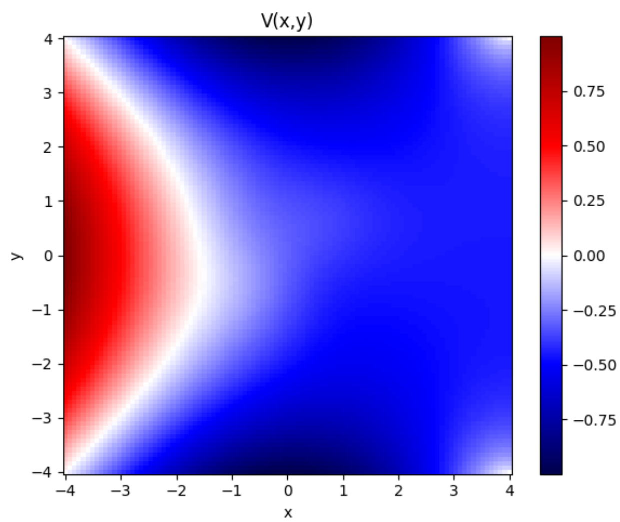

Figure 9: Potential for Task 4b with nonhomogeneous BC and $A_\rho = 1$.

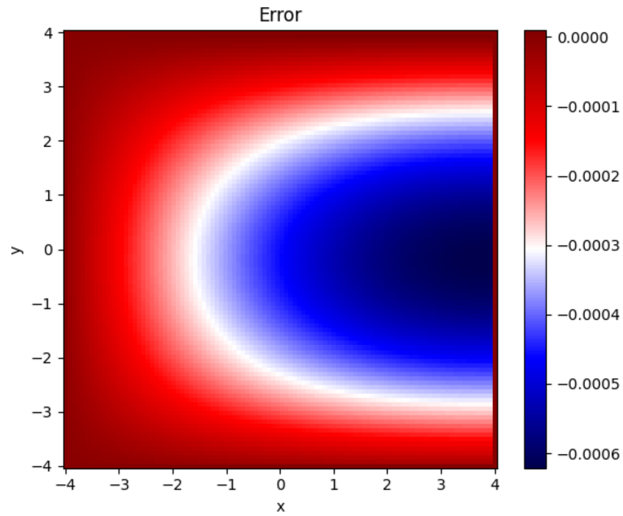

Figure 10: Error map for Task 4b.

4 Conclusions

In this project the two-dimensional Poisson equation was solved numerically using the Successive Overrelaxation (SOR) method. The implementation allowed us to investigate the influence of different boundary conditions, the presence or absence of charge density, and the role of the relaxation parameter ω on the convergence behavior.

The results confirmed that the potential distribution strongly depends on the applied boundary conditions when the charge density is zero, whereas for $A_\rho = 1$ the solution becomes dominated by the source term in the central region of the domain. Introducing a Neumann boundary condition on one edge produced the expected flattening of the potential near the boundary, demonstrating the method's flexibility in handling mixed BCs.

The convergence study showed that increasing the relaxation parameter significantly reduces the number of iterations required for stabilization, with ω values close to 2 providing the fastest convergence. Minor deviations and residual errors observed in some plots likely arise from incomplete convergence or implementation imperfections, but the overall qualitative agreement with theoretical expectations is clear.

Overall, the SOR method proved to be an efficient and intuitive approach for solving the Poisson equation on a discrete mesh, and the obtained results illustrate well the interplay between boundary conditions, charge distribution, and numerical acceleration.

References

[1] Dr. Hab. Eng. Chwiej, T. *Project: Solving Poisson equation with Successive Overrelaxation method (SOR)*. AGH University of Krakow. Retrieved from https://galaxy.agh.edu.pl/~chwiej/comp_phys/labs/5_poisson.pdf